

SONBONG, Korea, Democratic People's Republic of

WMO: 470030

Lat: 42.317N

Lon: 130.400E

Elev: 3

StdP: 101.29

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.8	(c) -13.3	(d) -26.6	(e) 0.3	(f) -10.7	(g) -24.4	(h) 0.4	(i) -10.0	(j) 14.9	(k) -12.4	(l) 13.3	(m) -10.5	(n) 5.3	(o) 320	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.6	(c) 27.7	(d) 23.1	(e) 26.2	(f) 22.6	(g) 25.0	(h) 22.1	(i) 24.5	(j) 26.3	(k) 23.6	(l) 25.3	(m) 22.8	(n) 24.4	(o) 2.7	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.8	25.6	23.0	17.8	24.8	22.2	16.9	24.0	74.3	26.2	70.9	25.4	67.8	24.4	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	8.9	7.5	-16.9	31.4	1.5	1.3	-18.0	32.3	-18.9	33.1	-19.7	33.9	-20.8	34.9
			-17.7	25.6	1.4	1.6	-18.7	26.8	-19.5	27.7	-20.2	28.6	-21.2	29.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-6.7	-4.7	0.5	6.6	11.4	15.8	20.1	21.9	16.9	10.4	2.0	-4.4		
	DBStd	10.18	3.35	3.46	3.37	3.02	3.10	2.77	2.81	2.18	2.71	3.17	4.34	3.67		
	HDD10.0	2073	518	412	294	109	19	0	0	0	0	33	241	446		
	HDD18.3	4139	777	646	553	351	216	85	14	2	58	244	490	704		
	CDD10.0	1184	0	0	0	8	63	176	313	368	208	47	1	0		
	CDD18.3	206	0	0	0	0	1	10	68	112	16	0	0	0		
	CDH23.3	740	0	0	0	1	5	41	204	441	48	0	0	0		
	CDH26.7	98	0	0	0	0	1	8	24	59	6	0	0	0		
(14) Wind	WSAvg	2.4	3.4	3.2	2.8	2.5	2.2	1.9	1.7	1.6	1.6	2.2	2.8	3.4		
(15) Precipitation	PrecAvg	791	7	10	15	45	91	115	173	149	99	52	26	10		
	PrecMax	1137	38	45	56	165	254	313	451	292	312	185	112	38		
	PrecMin	565	0	0	0	1	2	16	44	16	11	3	1	0		
	PrecStd	154	10	11	15	34	65	62	94	66	82	43	24	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	6.7	12.6	19.3	23.6	27.2	29.2	30.3	27.2	21.1	14.3	6.7		
		MCWB	0.3	2.3	5.8	10.3	14.8	19.1	23.5	24.0	20.1	15.1	10.1	3.8		
	2%	DB	1.6	4.1	9.6	16.1	20.7	24.0	27.0	28.1	24.5	19.0	12.2	4.3		
		MCWB	-1.1	0.7	4.2	9.2	13.9	18.5	23.6	23.4	19.4	14.9	8.7	1.5		
	5%	DB	0.2	2.5	7.6	13.9	18.5	21.7	25.4	26.8	23.0	17.7	10.5	2.6		
		MCWB	-2.2	-0.2	3.4	8.4	12.7	18.1	23.1	23.1	18.9	14.1	7.3	0.1		
	10%	DB	-1.3	1.1	5.9	11.8	16.5	20.0	24.2	25.6	21.9	16.4	8.6	1.1		
		MCWB	-3.5	-1.2	2.5	7.5	12.3	17.7	22.4	22.7	18.3	13.1	5.9	-1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	3.2	6.9	12.2	16.7	21.4	25.4	26.1	22.8	17.6	11.2	4.6		
		MCDB	2.9	5.5	11.0	16.7	20.7	24.3	27.1	28.2	25.2	19.2	13.4	6.2		
	2%	WB	-0.5	1.5	5.2	10.7	15.2	19.9	24.2	24.8	21.1	16.2	9.5	2.0		
		MCDB	1.2	3.3	8.2	14.6	19.0	22.3	25.8	26.9	23.0	18.0	11.5	3.8		
	5%	WB	-2.0	0.3	4.2	9.3	14.1	18.8	23.3	24.1	20.1	15.0	7.9	0.4		
		MCDB	-0.2	2.1	6.6	12.4	16.9	20.7	24.9	25.7	21.9	16.9	9.8	2.1		
	10%	WB	-3.3	-1.0	3.1	8.1	13.0	17.8	22.5	23.3	19.2	13.8	6.2	-0.9		
		MCDB	-1.5	1.0	5.4	10.8	15.2	19.4	24.0	24.9	21.0	15.8	8.5	0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	7.7	7.8	6.3	4.6	4.0	5.6	8.6	9.5	7.7	7.3		
		MCDBR	10.0	9.7	9.9	11.6	10.9	8.9	6.5	7.9	9.7	10.8	10.5	9.6		
	5% WB	MCWBR	8.2	7.5	6.3	6.4	5.5	4.8	4.1	4.7	5.9	7.9	8.0	7.9		
		MCWBR	10.1	9.3	9.1	9.9	8.8	6.3	5.4	5.7	7.6	9.5	10.0	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.404	0.478	0.567	0.602	0.596	0.556	0.503	0.446	0.452	0.406	0.358		
		taud	2.128	1.989	1.856	1.711	1.714	1.807	1.978	2.119	2.186	2.027	2.053	2.100		
	Ebn at Noon		759	764	751	715	704	711	736	765	780	705	674	698		
		Edh at Noon	103	140	181	226	231	211	176	149	129	134	112	98		
(44) All-Sky Solar Radiation	RadAvg		2.23	3.17	4.28	4.87	4.96	4.83	4.39	4.53	4.28	3.32	2.23	1.87		
	RadStd		0.16	0.20	0.25	0.39	0.54	0.78	0.60	0.38	0.37	0.23	0.24	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page